

Partitions, Invariants, and the Persistence of Distinguishable Structure

A Reduction of Distinction Guard Theory to a Defensible Mathematical Core

Abstract

A recurring observation across topology, quantum information, gauge theory, and non-equilibrium thermodynamics is that persistent physical structures are characterized not by their material constitution but by invariants that local dynamics cannot remove. This paper extracts and formalizes that observation. A distinction is defined as a partition operation on a state space, expressible equivalently as an indicator function, a projection operator, or a disjoint probability support. Persistence is defined as invariance of such a partition under the dynamics available to a system. Three known mechanisms by which partitions become dynamically invariant are reviewed: commutation of projectors with error operators in operator algebra, topological charge conservation in differential geometry, and conditional independence across a Markov blanket in probability theory. These three mechanisms are shown to instantiate a small number of independent protection principles, namely symmetry constraints, topological constraints, and informational constraints, each with established physical interpretations in quantum error correction, topological phases of matter, and stochastic thermodynamics respectively. The invariance condition is treated as binary throughout the main development; the relationship between this binary condition and the graded robustness of partitions under bounded perturbation is identified as an open extension. The paper does not propose a new ontology, a unification of these mechanisms, or any claim that distinction is ontologically prior to physical description. It identifies a defensible classification of existing persistence mechanisms and states the conditions under which a stronger unification claim would need to be tested.

1. Introduction

Across several areas of physics, a similar pattern recurs. A system contains some structure, label, or configuration that distinguishes it from its surroundings or from an alternative configuration. Under generic dynamics, such distinctions tend to be erased, in the sense that the system relaxes toward a state in which the distinction no longer affects observable behavior. In certain cases, however, the distinction is not erased, even under dynamics that would erase a generic distinction. Topological solitons retain their winding number under smooth local perturbations. Logical qubits in a quantum error-correcting code retain their encoded information under local noise. Systems separated by a Markov blanket retain conditional independence from their environment under fluctuation.

In each of these cases, physicists do not appeal to a new physical substance to explain the persistence. They appeal to a mathematical constraint, an invariant, a conserved charge, a commutation relation, or a conditional independence structure, that the dynamics cannot violate. This paper asks a narrow question: can the common structure underlying these constraints be stated in a single, minimal mathematical vocabulary, without making any claim about what is ontologically fundamental.

The motivation for this reduction is methodological. A manuscript that originated this analysis proposed a broad classification scheme, termed Distinction Guard Theory, organized around the idea that “distinctions” are a fundamental category and that seven classes of “guards” protect them, embedded within a larger cosmological framework concerning the emergence of spacetime from an informational substrate. That broader framework is not pursued here. What

survives the reduction is considerably smaller: a description of partition operations and their invariance properties, expressed in standard mathematical language, that recovers known results in quantum error correction, topological phases, and stochastic thermodynamics as special cases. The Discussion section addresses what would be required to support the larger claims, and why those claims are not addressed in the main text.

2. Mathematical Preliminaries

Let a system be described by a state space X , which may be a set, a measurable space, a Hilbert space, or a manifold depending on context. A partition of X is a decomposition into disjoint, jointly exhaustive subsets, subspaces, or supports.

Three equivalent representations of a partition are used throughout.

Indicator functions.

For a measurable space X and a subset $A \subseteq X$, the indicator function $\chi_A : X \rightarrow \{0, 1\}$ takes value 1 on A and 0 elsewhere. A collection of disjoint sets $\{A_i\}$ with $\cup A_i = X$ defines a partition, and the corresponding indicator functions satisfy $\sum \chi_{A_i}(x) = 1$ for all $x \in X$.

Projection operators.

For a Hilbert space $\mathcal{H}$, a set of projectors $\{\hat{P}_i\}$ satisfying $\hat{P}_i \hat{P}_j = \delta_{ij} \hat{P}_i$ and $\sum \hat{P}_i = \hat{I}$ partitions $\mathcal{H}$ into mutually orthogonal subspaces. This is the quantum-mechanical analogue of the indicator function decomposition.

Disjoint probability supports.

For a probability density $p(x)$ on X , a partition into disjoint supports corresponds to a decomposition of the sample space such that $p(x)$ restricted to each region defines a conditional distribution, and the regions are mutually exclusive in the sense that an outcome belongs to exactly one region.

These three representations are mathematically related: a projector defines an indicator function on the spectrum of an observable, and a probability density restricted to disjoint supports is the classical limit of a diagonal density matrix with projector decomposition. The paper uses the term **distinction** to refer to any such partition, with the understanding that this is a definitional choice rather than a substantive claim. A distinction in this sense is an operator or a set-theoretic structure, not an object, a substance, or a fundamental constituent of reality.

The arithmetic statement $1 + 1 \neq 1$ is sometimes invoked informally to express the idea that combining two distinct systems does not yield a single system identical to either original. In the language above, this is simply the statement that if $A \cap B = \emptyset$ and both are nonempty, then $A \cup B \neq A$ and $A \cup B \neq B$, or equivalently that $\mathbb{R}^1 \oplus \mathbb{R}^1 = \mathbb{R}^2 \neq \mathbb{R}^1$ for independent one-dimensional subspaces. This is a true and elementary statement about set unions and direct sums. It does not by itself imply anything about cosmological irreversibility, the impossibility of the universe returning to a uniform state, or the necessity of ongoing structural generation. Those further claims would require additional physical assumptions connecting dimension counting in an abstract state space to the thermodynamic or cosmological evolution of a physical system, and are not established here. This point is revisited in the Discussion.

3. Distinctions as Partition Operations

A distinction, as defined in Section 2, is a partition of a state space into disjoint regions, subspaces, or supports. This section makes explicit what is and is not claimed by this definition.

The definition is purely structural. It does not assert that distinctions are fundamental, that reality is “made of” distinctions, or that partition operations exist independently of the systems they describe. A partition is a mathematical description applied to a system; it is not itself a physical entity. This is the same status that a coordinate system, a basis choice, or a measure has in physics: indispensable for description, but not ontologically privileged.

What the definition does provide is a common vocabulary for comparing structures that arise in different subfields under different names. A topological defect, a logical subspace of an error-correcting code, and a thermodynamic system separated from its environment by a boundary are, at the level of this vocabulary, all instances of a partition of some underlying space into regions with different properties. Whether that partition persists under the dynamics relevant to each case is the subject of Section 4.

4. Invariant-Preserving Structures

4.1 Formal definition

Let X be a state space with a partition $\{A_i\}$, and let T denote the dynamics available to the system, whether a unitary evolution, a stochastic process, a renormalization group flow, or a more general map on X . The partition is *invariant under T* if T maps each region A_i to itself, or more generally if the labeling of which region a state belongs to is preserved by T for all states and all times, up to a set of dynamics-dependent exceptions of measure zero or energy cost diverging in an appropriate limit.

A partition that is invariant under the dynamics available to a system is called persistent. A partition that is not invariant is subject to erasure, meaning that the dynamics can map a state in one region to a state in another region, so that the original partition no longer corresponds to a meaningful distinction after evolution.

4.2 Relationship to persistence

This definition recovers the standard physical notion of a conserved quantity or protected subspace as a special case. A conserved charge Q defines a partition of phase space into level sets of Q , and conservation of Q under the dynamics is exactly invariance of this partition. A logical subspace of a quantum error-correcting code defines a partition of the full Hilbert space into the code space and its complement, and the code's protection against a noise channel is invariance of this partition under that channel, up to correctable errors. A system separated from its environment by a Markov blanket defines a partition into internal and external degrees of freedom, and the persistence of this separation under the joint dynamics is invariance of the partition in the sense above.

In each case, the question “why does this distinction survive?” reduces to the question “what property of the dynamics T guarantees invariance of this particular partition?” Section 5 reviews the three principal answers to this question that appear in established physics.

5. Taxonomy of Protection Mechanisms

Three mechanisms by which a partition can be invariant under dynamics are well established in different subfields. Each is stated here in its native mathematical language, with the relationship between them noted explicitly.

5.1 Symmetry constraints (the commutation mechanism)

Let a partition of a Hilbert space $\mathcal{H}$ be defined by projectors $\{\hat{P}_i\}$, and let the dynamics include a set of operators $\{\hat{E}_k\}$ representing local noise or interaction terms (for example, Lindblad or Kraus operators). The partition is invariant under this dynamics if

$$[\hat{P}_i, \hat{E}_k] = 0 \quad \forall i, k$$

When this condition holds, the noise operators cannot move a state across the boundaries of the partition; they can act within each subspace but cannot change which subspace a state occupies. This is the operating principle of quantum error-correcting codes with respect to correctable error sets, and more generally of any conserved quantity whose associated projectors commute with the Hamiltonian or with the relevant noise operators. This mechanism is classified here as a symmetry constraint, since commutation with the dynamics is the algebraic statement of a conserved quantity associated with a symmetry of the dynamics, by Noether's theorem in its operator form.

5.2 Topological constraints (the invariant-charge mechanism)

Let the system be described by a connection or field configuration on a fiber bundle over a base space, and let a partition be defined by the value of a topological charge

$$Q = \int_{\Sigma} dA = \oint \partial \Sigma A$$

for some closed differential form dA and compact subspace Σ . Because $d^2 = 0$, local smooth deformations of the field configuration cannot change Q ; Q is invariant under any dynamics that proceeds through smooth, local field configurations. Changing Q requires a discontinuous or non-local change to the field configuration, which in physical systems corresponds to an energy cost that diverges in an appropriate limit (for example, the energy required to unwind a soliton's winding number across an infinite system). This mechanism is classified here as a topological constraint. It underlies the protection of solitons, anyons, and the topologically ordered phases of matter recognized by the 2016 Nobel Prize in Physics for work on topological phase transitions.

5.3 Informational constraints (the conditional-independence mechanism)

Let the state space be a probability manifold with density $p(x)$, and let a partition divide the degrees of freedom into internal states i , external states e , and a boundary or blanket b . The partition is invariant in the sense of maintaining a conditional independence structure if

$$\forall x \quad p(i, e | b) = 0 \Rightarrow p(i, e | b) = p(i | b) p(e | b)$$

When this condition holds, fluctuations in the external states cannot directly induce changes in the internal states except through the boundary variables; the internal and external partitions remain statistically distinguishable. This is the defining structure of a Markov blanket in stochastic and statistical descriptions of systems with an effective boundary, and is the relevant mechanism in non-equilibrium steady states where a system maintains a distinguishable internal configuration against a fluctuating environment. This mechanism is classified here as an informational constraint, following its standard description in terms of conditional independence and information geometry.

5.4 Relationships among the three mechanisms

These three mechanisms are not claimed to be ontologically distinct kinds of protection occurring in different “substances.” They are three mathematical settings, operator algebra,

differential geometry, and probability theory, in which the same underlying requirement appears: the dynamics must fail to connect the interior of a region to its exterior. Whether this failure is expressed as a vanishing commutator, a topologically protected integral, or a vanishing conditional gradient is determined by which mathematical description is natural for the system at hand, not by a difference in the underlying physical principle.

It is plausible, and is suggested by existing research programs reviewed in Section 6, that all three are related through information geometry, in the sense that each corresponds to a direction on an appropriate parameter manifold along which the relevant metric (a Fisher information metric, in the informational case, or an analogous geometric quantity in the other two cases) becomes singular or the associated relaxation time diverges. This is stated here as a plausible extension rather than an established result, since a general proof that all topological and symmetry-protected invariants can be expressed as singularities of a single information-geometric structure is not, to this author's knowledge, established in the literature in full generality. Existing results establish this correspondence in specific cases, for example the relationship between topological order and entanglement entropy, or between symmetry and Fisher information in quantum metrology, but a fully general unification is an open research question rather than a settled result.

No further taxonomic classes are proposed. The original manuscript's seven-category taxonomy, topological, informational, symmetry, dynamical, constraint, historical, and path guards, is addressed in the Reduction Log; the dynamical, constraint, historical, and path categories are found to be either special cases or redescrptions of the three mechanisms above, rather than independent protection principles.

6. Relationship to Existing Frameworks

Information theory

The partition framework above is consistent with, and in the conditional-independence case directly drawn from, standard information-theoretic descriptions of correlated systems. No claim is made that information is more fundamental than the systems it describes; the framework treats information-theoretic quantities such as entropy, mutual information, and conditional independence as properties of partitions and distributions, in line with their standard usage.

Quantum error correction

Section 5.1 restates, in general form, the condition for a code space to be protected against a given error set. The framing of “distinction guards” as commuting projector algebras is a direct restatement of the Knill-Laflamme conditions and their generalizations, not a new result.

Gauge theory

Section 5.2's topological charge mechanism is standard in the description of solitons, instantons, and topological defects in gauge theories. The “It from Qubit” research program, associated with work by Maldacena, Preskill, and collaborators on the relationship between spacetime geometry and quantum entanglement structure, is an active area in which informational and topological protection mechanisms are studied together; this paper's taxonomy is compatible with, but considerably narrower than, that program, and makes no claims about emergent spacetime.

Non-equilibrium thermodynamics

Section 5.3's Markov blanket formulation is used in the free-energy and active-inference literature to describe systems maintaining a distinguishable internal state against environmental fluctuation. The relevant claims here are restricted to the existence and mathematical form of the conditional independence condition, not to any broader claims about the role of such systems in cosmological or biological generation of complexity.

Information geometry

The suggestion in Section 5.4 that all three mechanisms may correspond to singularities of an information-geometric structure connects to the broader information geometry literature, but is flagged explicitly as an open question rather than a result of this paper.

7. Discussion

7.1 Philosophical implications

The reduction performed in this paper removes any claim that “distinctions” are fundamental constituents of reality, that information is ontologically prior to physical description, or that the universe is “composed of” partitions. These were claims present in the source manuscript's broader framework and are not supported by the mathematics retained here. What remains is a classification of known invariance mechanisms, stated in a common vocabulary. The philosophical question of whether this common vocabulary indicates something deep about the structure of physical description, or is simply a convenient unification of notation, is open and is not adjudicated here.

7.2 The status of “ $1 + 1 \neq 1$ ” as an organizing claim

The source manuscript treated the inequality $1 + 1 \neq 1$, in its set-theoretic and linear-algebraic forms, as what it termed “the ultimate axiom” of the framework, from which claims about the irreversibility of cosmological expansion and the necessity of ongoing generation of complexity were derived. As stated in Section 2, the inequality itself is a correct and elementary mathematical fact about disjoint unions and direct sums of independent spaces. The inference from this fact to claims about physical irreversibility or cosmological necessity is an analogy rather than a derivation: it requires an additional, unstated assumption that the dimension or cardinality of an abstract description space tracks a physically conserved or monotonically increasing quantity in the actual universe. No such assumption is justified here, and the inferential step is therefore not retained in the main text. It is recorded as a speculative interpretation that may be of interest for further work but is not part of the reduced framework.

7.3 Open problems

Two open problems are identified by this reduction, both already present in the literature rather than novel to this paper. First, whether the three protection mechanisms of Section 5 admit a single information-geometric description, as suggested in Section 5.4, remains open in general. Second, whether the classification of Section 5 is exhaustive, that is, whether every known mechanism of dynamical invariance for a partition can be assigned to one of the three categories or a combination of them, has not been verified against the full range of known protected phases and codes, and would require a systematic survey beyond the scope of this paper.

7.4 Limitations

This paper makes no claims about emergence of spacetime, geometry, or physical law from a more fundamental substrate. It makes no claims about cosmology. It does not propose a new physical theory, a new fundamental force, or a new ontology. It is a classification and notational unification of existing results, intended to clarify the relationship between persistence phenomena studied in different subfields under different names.

7.5 Binary versus graded invariance

The invariance condition in Section 4 is stated as a binary property: a partition is either preserved by the dynamics T , or it is not. This is appropriate for the topological and symmetry mechanisms of Sections 5.1 and 5.2, where the relevant quantities are either exactly conserved by an algebraic or topological identity or they are not, there is no intermediate case for a commutator that vanishes “mostly” or a topological charge that is “approximately” quantized.

Many persistent structures, however, are not protected in this all-or-nothing way. A logical qubit's protection degrades continuously with noise strength below threshold rather than switching off discontinuously. A Markov blanket's conditional independence can hold approximately, with a small residual coupling between internal and external states, rather than exactly. More generally, for a given partition and a given dynamics, there is often a range of perturbations or deformations under which the partition continues to be preserved, and the size of this range is itself a meaningful physical quantity, distinct from the yes-or-no question of whether the partition survives any perturbation at all.

The framework developed here does not provide a vocabulary for this graded notion of robustness. A natural extension would characterize, for a given partition and transformation set, the measure or extent of the region of perturbation-space under which invariance continues to hold, rather than only the binary fact of invariance under a fixed, exactly specified dynamics. This would allow comparison between, for example, two partitions that are both preserved under their respective dynamics but differ in how much deformation each can tolerate before the distinction ceases to be meaningful. Developing such a measure, and relating it to existing notions such as error thresholds in quantum error correction or stability margins in dynamical systems, is left as an open extension and is not attempted here.

8. Conclusion

A distinction, defined as a partition of a state space expressible as an indicator function, projection operator, or probability support, persists when it is invariant under the dynamics available to the system. Three established mechanisms produce such invariance: commutation of the partition's projectors with the dynamics (a symmetry constraint), conservation of a topological charge under smooth local dynamics (a topological constraint), and conditional independence across a boundary under stochastic dynamics (an informational constraint). These three mechanisms recover, respectively, the protection condition for quantum error-correcting codes, the protection of topological phases and solitons, and the persistence of systems with a Markov blanket in non-equilibrium thermodynamics. Whether these three mechanisms reduce further to a single information-geometric principle is an open question. The framework as developed treats invariance as binary; extending it to a graded measure of robustness, as discussed in Section 7.5, is left for future work. No claim beyond this classification is made.

Appendix: Reduction Log

The following table records each major claim or concept from the source manuscript, its reduced form in this paper, and the reason for the reduction, following the five passes specified in the audit procedure.

Original Claim	Reduced Form	Reason for Reduction
"Information Sea," geometrogenesis, and the Informational Manifold Hypothesis as the cosmological substrate from which spacetime emerges	Removed entirely	Unfalsifiable in current form; no mathematical structure was specified beyond analogy to existing emergence-of-spacetime programs (Category D)
"Distinction" as a fundamental ontological category, prior to physical description	A distinction is a partition operation (indicator function, projector, or probability support) on a state space	The ontological claim is philosophical and unfalsifiable; the structural claim is established mathematics (Pass 5: weakest defensible statement)
Seven guard classes: Topological, Informational, Symmetry, Dynamical, Constraint, Historical, Path	Three protection mechanisms: symmetry (commutation), topological (invariant charge), informational (conditional independence)	Dynamical and Constraint guards redescribe the symmetry and topological mechanisms; Historical and Path guards describe consequences of persistence rather than independent protection mechanisms (Pass 4)
"Reconciliation Dynamics (Γ)" as a universal interaction layer governing all distinction interactions	Removed as a named framework; retained implicitly as "the dynamics T" in Section 4, with no universal claims about its form	No general mathematical form for Γ was specified; as stated, the claim redescribes ordinary equilibration dynamics (Pass 2)
TNDS persistence variable S° and I_6 compositional fertility as primary quantities of a unified framework	Removed; persistence treated via the invariance condition of Section 4 without new named variables	No independent definition of S° or I_6 in terms of measurable quantities was given; these function as placeholders for persistence and generation rates already covered by standard dynamical systems notation (Pass 1)
"Four primitives of reality": Distinction, Interaction, Persistence, Generation, claimed to underlie all physical systems from qubits to galaxies	Distinction and Persistence retained as defined objects (Sections 3-4); Interaction retained implicitly as "the dynamics"; Generation removed entirely	The first three redescribe partitions, dynamics, and invariance already covered by standard formalism. "Generation" as stated has no specified mechanism or falsification condition (Category D, removed)
$1 + 1 \neq 1$ as "the ultimate axiom" implying the universe cannot run backward to a uniform point and is "mathematically forced to keep expanding"	Retained as an elementary mathematical fact about disjoint unions and direct sums (Section 2); the cosmological inference removed and flagged as analogy (Section 7.2)	The arithmetic statement is correct (Category A). The cosmological inference adds unstated physical assumptions and is an analogy, not a mechanism (Pass 3: Category C, moved to Discussion)
"Guarded distinctions create historical memory in the state space," connecting persistence, irreversibility, topology, information, and complexity	Removed as a standalone claim; the underlying observation is implicit in Section 4's definition of invariance	As stated, this restates the definition of an invariant in dynamical systems using new vocabulary without adding content (Pass 2)

Three formal definitions of a Distinction Guard: commutation $[\hat{P}_i, \hat{E}_k]=0$; topological charge $Q=\int \sum dA$; Markov blanket $\forall x p(i,e b)=0$	Retained as the core of the paper (Section 5.1-5.3)	Correct restatements of established results in quantum error correction, gauge theory/topology, and stochastic thermodynamics (Category A); this is the load-bearing content of the manuscript
Claim that “anisotropy on the information manifold” / a singular Fisher information metric is the common denominator of all guard types, answering the open question affirmatively	Reduced to a plausible extension (Category B), stated as an open question rather than an answered one	A general proof of this unification was not provided; specific cases are known, but the general claim is not established (Pass 3: moved from “answer” to “open problem”)
“It from Qubit,” topological phase transitions (2016 Nobel Prize), and Markov blanket / active inference literature cited as placing the framework “in the exact same domain” as these programs	Retained as relevant background (Section 6), reframed as literature the reduced framework is compatible with and narrower than	The citations are accurate as pointers to established physics, but were originally used to lend authority to the full speculative framework rather than the narrow core that actually overlaps with them (Pass 3: scope correction)